Geophys. J. Int. (1990) **103**, 85–94

Palaeomagnetic study of eastern Tibet—deformation of the Three Rivers region

Y. Otofuji,[1] Y. Inoue,[1] S. Funahara,[1] F. Murata[1] and X. Zheng[2]

[1] *Department of Earth Sciences, Faculty of Science, Kobe University, Kobe 657, Japan*

[2] *Institute of Geology, Academia Sinica, Beijing, China*

Accepted 1990 April 24. Received 1990 April 24; in original form 1989 November 20

SUMMARY

A palaeomagnetic study was undertaken for Cretaceous red beds and granitic rocks of three geological provinces (the Lhasa, the Qiangtang and the Sichuan) along a road between Chengdu and Lhasa in eastern Tibet in an attempt to determine aspects of deformation of SE Asia. High-temperature characteristic directions were isolated from 20 sites. The tilt corrected data from the Qiangtang Terrane at Markam (29.7°N, 98.6°E) exhibit largely clockwise deflected declinations with steep inclinations ($D = 48.2°$, $I = 49.0°$, $\alpha_{95} = 8.8°$), whereas the Lhasa Terrane at Basu (30.1°N, 96.9°E) and the Sichuan province at Ya'an (30.1°N, 103.0°E) have northerly directions with shallow inclinations ($D = -3.9°$, $I = 23.4°$, $\alpha_{95} = 19.2°$ and $D = 2.1°$, $I = 30.7°$, $\alpha_{95} = 11.3°$, respectively). The palaeomagnetic data indicate that the Qiangtang Terrane was subjected to clockwise rotation through 50° with respect to the neighbouring provinces of the Lhasa and Sichuan since the Cretaceous. Tectonic deformation occurred later than the Cretaceous in the eastern part of the Qiangtang Terrane and formed the arcuate shape of the orogenic belt in the Three Rivers region.

Key words: collision, deformation, eastern Tibet, palaeomagnetism, tectonics.

INTRODUCTION

It has become apparent in the last few years that the post-Eocene convergence of India with Asia has given rise to diffuse deformation in Asia (Molnar & Tapponnier 1978; Allègre *et al.* 1984; Patriat & Achache 1984; Achache, Courtillot & Xiu 1984; Chang *et al.* 1986). Elucidating aspects of deformation of Asia are important for deciphering the tectonic history of Asia and for understanding the dynamical character of a non-plate-like body of the continental lithosphere. Since the Tibetan region has constituted the southern margin of Asia prior to the collision of India, it provides a magnificent opportunity to study the deformation of the Asian continent.

A peculiar kink shape is observed in the geomorphology at the Three Rivers region in eastern Tibet, which is bounded on the west by the Tibetan Plateau and on the east by the Sichuan Basin (Fig. 1). Three large rivers, the Saluween, the Mekong and the Yangtze, flow southeastward in the Tibetan Plateau. These rivers change their flow direction to a north–south direction and concentrate in the narrow zone with a width of about 100 km in eastern Tibet. The large kink shape in this area has been interpreted as a reflection of a large tectonic deformation of the Asian continent (Molnar & Tapponnier 1975; Wang & Chu 1988).

We made an expedition to traverse eastern Tibet along the latitude of about 30°N from Lhasa to Chengdu for 2800 km (Fig. 1). More than 150 oriented samples were collected from Cretaceous red beds and granitic rocks in an attempt to detect intracontinental deformation in eastern Tibet from the palaeomagnetic point of view.

GEOLOGICAL SETTING AND SAMPLING

Eastern Tibet is a composite of four geological Terranes (Wei & Tan 1979; Zhang *et al.* 1984): from northeast to southwest, Yangtze, Bayan Kala, Qiangtang and Lhasa (Gamnian) (Fig. 1). According to an outline given by Allègre *et al.* (1984), Chang *et al.* (1986), Zhang *et al.*, (1984) and Zhao & Coe (1987), the evolution of Tibet is characterized as the successive accretion of several geological terranes to the active southern margin of the Asian continent since late Palaeozoic. The Yangtze Terrane began to collide with the SE rim of Asia in early Triassic, and their coalescence was completed during the Jurassic. The Qiangtang Terrane was sutured to the Bayan Kala Terrane in the west and to the Yangtze Terrane in the east along the Jinsha suture zone about 200 Myr ago. Around 120 Myr ago, the Lhasa Terrane accreted to the Qiangtang Terrane along the Banggong–Nujiang suture zone. The last

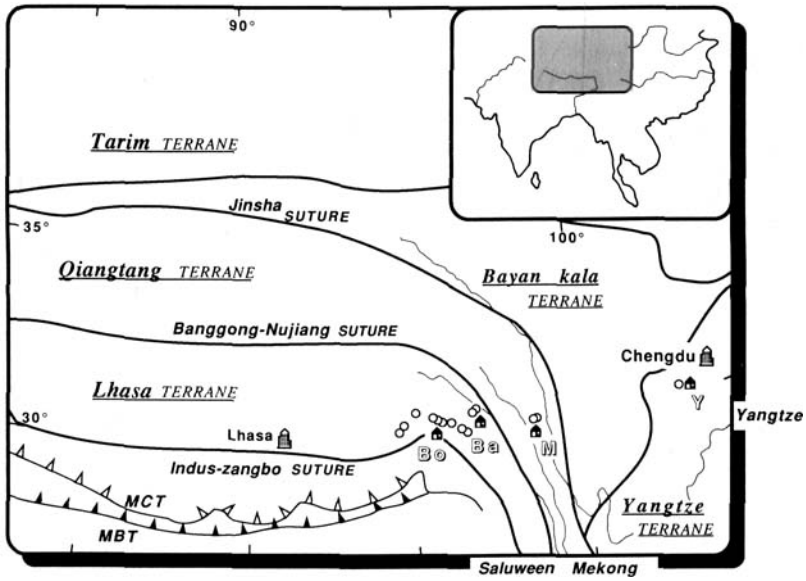


Figure 1. Sketch map of the major geologic terranes in eastern Tibet, separated by suture zones (heavy lines). Barbed lines are continental thrust (MCT, Main Central Thrust; MBF, Main Boundary Fault). Sampling localities are indicated by open circles along the sampling route from Lhasa, through Bomi (Bo), Basu (Ba), Markam (M) and Ya'an (Y) to Chengdu.

accretion occurred at about 50 Ma: the Indian continent moved northward and collided with the Lhasa Terrane.

Samples were collected at 35 sites from four areas in three geological terranes (see Fig. 1); Ya'an (nine sites) in the Yangtze Terrane, Markam (five sites) in the Qiangtang Terrane, and Basu (seven sites) and Bomi (14 sites) in the Lhasa Terrane. Samples consist of Cretaceous red sandstones of 19 sites and Cretaceous granites of 16 sites.

Ya'an

The Cretaceous system is widely distributed, with well-understood stratigraphy and good faunal control, in the Ya'an area which is located on the Sichuan province in the west of the Yangtze Terrane (Geological Map of the Tibetan Plateau 1981; Sichuan Institute of Geology and Mineralogy 1982; Hao *et al.* 1986). Samples were collected from three Cretaceous formations: the Tianmashan Formation, the Jianguan Formation and the Guankou Formation, in ascending order. The Cretaceous formations are mainly composed of purple-coloured sandstones and siltstones, and are exposed on both limbs of three synclinal structures in the Ya'an area. A total of 27 samples of nine sites come from both limbs, so that there is opportunity for a fold test.

Markam

The Markam area is one of the rare areas on the eastern part of the Qiangtang Terrane where the Cretaceous system (the Laoran Formation) is exposed (Geological Map of the Tibetan Plateau 1981; Hao *et al.* 1986), although the Jurassic strata are widely distributed in this terrane. Since many fossils of vertebrates (e.g. *Monkonosaurus lanwlacus Chao*) have been discovered, the Laoran Formation is assigned to a Barremian ~ Albian (Early Cretaceous) age (Hao *et al.* 1986). 15 samples of five sites were collected from purple-coloured sandstones of the Laoran Formation with a homocline structure dipping east at 46°–58° at the Markam.

Basu

The Cretaceous system (the Basu Formation) is exposed at the Basu area which is located on the eastern part of the Lhasa Terrane (Geological Map of the Tibetan Plateau 1981; Chen *et al.* 1986). The structure of this formation is a homocline dipping northeast at 46°–63° in the Basu area. The Basu Formation is composed of fossiliferous light red-coloured sandstones: fossils of bivalvia (*Trigonioides–Pseuduniotibetica*) have been assigned to the Albian–Aptian (Early Cretaceous) (Chen *et al.* 1986). A major NW–SE trending belt of plutonic rocks is exposed to the south of Basu (Geological Map of the Tibetan Plateau 1981). Ages for these granitic rocks were given from 128 to 72 Myr by the K–Ar method and from 127 to 69 Myr by the U–Pb method (So *et al.* 1982). 15 samples of five sites and 10 samples of two sites were collected from the Basu Formation and granitic rocks, respectively.

Bomi

The NW–SE trending plutonic belt is exposed at the Bomi area (Geological Map of the Tibetan Plateau 1981). Cretaceous to Miocene ages (20–95 Myr) were assigned to the granitic rocks exposed at the Basu area by the U–Pb and the K–Ar study (So *et al.* 1982). We collected 65 samples at 14 sites from these granitic rocks.

The block samples were oriented using a magnetic compass orientation device. Samples were distributed over distance ranging up to 10 m in each site. The present declination value at the sampling point was estimated from the calculation using the International Geomagnetic Reference Frame of 1986 (IAGA Division I Working Group 1 1987).

PALAEOMAGNETISM

Palaeomagnetic measurements

In the laboratory, individual specimens, 25 mm in diameter and 25 mm long, were prepared from each sample. Natural remanent magnetizations (NRMs) were measured with a ScT cryogenic magnetometer. All specimens were subjected to stepwise thermal demagnetization, up to 710 °C in the case of the Cretaceous red sandstones and 600 °C in the case of the granitic rocks. Red sandstones samples were also treated in alternating magnetic field (AF) up to 95 mT. Thermal demagnetization seemed to be the most efficient process for cleaning.

Zijderveld component plots (Zijderveld 1967) and equal area projections were used to assess the directional stability and coercivity spectrum of each specimen.

Palaeomagnetic results

The results are summarized in Tables 1 and 2. Principal component analysis (Kirschvink 1980) was used to estimate the direction represented by the collinear array of three or more data points defining the component of magnetization. A mean direction for each site was calculated by assigning unit weight to a specimen in the case of the Cretaceous red sandstones and to a sample in the case of the granitic rocks. We arbitrarily discarded results from any site for which the circle of 95 per cent confidence exceeded 30°.

Ya'an

Red sandstones from all nine sites have two-component magnetization (Fig. 2). Thermal treatment isolates two components, although AF demagnetization is not effective. The low blocking temperature (LT) component is destroyed between 100 °C and 530 °C. The LT component falls close to the present dipole field direction before correction of tilt (Fig. 3). Upon applying a tilt correction, the grouping of the LT components is fairly dispersed, which is a clear indication that the LT component was acquired after folding.

The high-temperature (HT) components appear in all

Table 1. *N* is the number of field-oriented samples, *n* is the number of specimens, α_{95} is the confidence circle, and *k* is the precision parameter. A mean direction for each site is calculated by assigning unit weight to specimens in the case of the sandstones and to a sample in the case of the granitic rocks. A mean direction is computed after Kirschvink (1980) in the range of unblocking temperature where the characteristic direction has been identified. UNI = single component; HT = high-temperature component; LT = component destroyed at low temperature. Mean direction for each formation and granitic body is also calculated and represented as the value at the representative point. * Number of sites for mean calculation.

Terrane	Site	Component	(°C)	N	(n)	Uncorrected Dec. (°)	Inc. (°)	α_{95} (°)	K	Corrected Dec. (°)	Inc. (°)	α_{95} (°)	K	Locality Lat. (°) (N)	Long. (°) (E)
Sichuan province	Ya'an	(Cretaceous red sandstones)													
	76	HT	0-680	3	5	-1.5	40.0	14.4	29.0	-6.2	32.3	14.4	29.0	30.2	102.9
		LT				no characteristic direction									
	77	HT	100-710	3	6	21.6	37.2	10.4	42.9	0.2	27.1	10.4	42.6	30.1	102.9
		LT				no characteristic direction									
	78	HT	0-530	3	6	52.3	57.0	18.7	13.7	10.1	45.2	18.8	13.6	30.1	102.9
		LT				no characteristic direction									
	79	HT	400-710	3	5	33.2	22.4	8.2	87.3	19.6	29.7	8.3	86.6	30.0	102.9
		LT	0-450	3	5	-16.9	60.3	10.3	56.0	-43.2	40.9	10.2	57.2		
	80	HT	100-710	3	6	-16.5	33.7	26.8	7.2	-21.3	27.7	26.8	7.2	30.1	103.1
		LT	0-530	3	5	-15.9	50.5	23.4	11.6	-24.5	44.3	23.2	11.8		
	81	HT	250-680	3	4	-176.9	-34.2	25.9	13.6	-158.7	-34.4	25.9	13.6	30.1	103.1
		LT	0-450	3	6	59.3	41.3	23.7	8.9	69.1	20.7	23.7	8.9		
	82	HT	0-700	3	5	7.2	36.4	11.7	43.6	21.1	39.1	11.8	43.1	30.1	103.1
		LT	0-400	3	5	20.2	73.1	23.1	11.9	67.2	66.1	23.1	11.9		
	83	HT	250-700	3	6	179.1	-1.3	5.8	135.0	-177.9	-15.9	5.8	134.6	30.0	103.0
		LT	0-350	2	3	12.4	68.8	18.5	45.3	73.4	66.0	18.7	44.5		
	84	HT	200-700	3	6	157.7	2.5	5.8	134.6	158.9	-15.4	5.8	135.4	30.0	103.0
		LT	0-150	3	6	-16.3	39.7	12.0	32.3	0.5	54.5	12.0	32.1		
	mean	HT		9*		5.9	30.8	17.8	9.3	2.1	30.7	11.3	21.7	30.1	103.0
		LT		6*		5.9	59.5	19.6	12.7	19.2	59.8	32.7	5.2		
Qiangtang Terrane	Markam	(Barremian /Albian red sandstones)													
	70	HT	0-680	3	6	-30.2	53.9	4.7	208.8	47.3	49.2	4.6	211.0	29.7	98.6
		LT				no characteristic direction									
	71	HT	250-710	3	6	-22.2	53.6	15.3	20.0	51.8	56.0	15.3	20.1	29.7	98.6
		LT				no characteristic direction									
	72	UNI	0-710	3	6	-30.8	54.4	6.0	127.5	54.8	54.3	6.0	128.0	29.7	98.6
	73	UNI	0-710	3	6	-23.8	56.2	6.2	118.7	56.2	42.9	6.2	119.7	29.7	98.6
	74	HT	100-710	3	6	-16.4	44.5	6.1	123.3	33.3	40.9	6.0	124.0	29.7	98.6
		LT				no characteristic direction									
	mean	HT+UNI		5*		-24.3	52.7	5.6	187.0	48.2	49.0	8.8	76.9	29.7	98.6

Table 1. (*continued*)

Lhasa Terrane														
Basu	(Albian/Aptian red sandstones)													
63	UNI	0-700	3	6	-29.1	63.8	4.8	196.9	7.7	12.1	4.8	200.1	30.1	96.9
64	UNI	0-700	3	6	-71.2	66.9	7.6	79.6	9.0	30.5	7.6	79.3	30.1	96.9
65	HT				no characteristic direction.									
	LT	0-350	3	5	7.2	43.1	17.9	19.2	9.3	-2.6	18.0	19.0	30.1	96.9
66	HT	250-710	2	4	-85.5	68.7	10.0	85.1	19.1	60.0	10.0	85.4	30.1	96.9
	LT	0-300	3	9	13.1	39.8	9.3	31.4	22.4	3.6	9.3	31.4		
67	UNI	0-680	3	5	-24.9	52.8	17.8	19.4	-6.1	13.2	17.9	19.3	30.1	96.9
mean	HT+UNI			4*	-47.8	65.4	17.1	29.7	3.3	29.0	28.0	11.7	30.1	96.9
Basu	(granitic rocks : 69Ma~127Ma[U-Pb] and 72Ma~128Ma[K-Ar])													
61	HT	200-590	4	4	-18.7	10.6	8.6	114.8					30.0	96.7
	LT				no characteristic direction									
62	HT	250-590	4	4	166.2	-13.3	4.5	416.6					30.0	96.7
	LT				no characteristic direction									
mean				2*	-16.3	12.0							30.0	96.7
Bomi	(granitic rocks : 65Ma~95Ma [U-Pb] and 20Ma~38Ma [K-Ar])													
46	UNI				no characteristic direction									
47	UNI	0-350	3	3	40.1	42.7	13.6	83.7					29.7	94.7
48	UNI	0-150	4	4	-58.9	65.0	19.9	22.3					29.8	94.7
50	UNI	0-620	5	5	12.5	50.2	10.7	51.8					29.9	95.7
51	UNI	0-620	5	5	10.9	31.1	18.2	18.7					29.9	95.8
52	UNI	0-530	5	5	-4.1	47.7	20.6	14.7					29.8	95.9
53	UNI	0-500	3	3	19.6	44.7	26.9	22.1					29.8	95.9
54	UNI	0-200	5	5	18.3	60.0	20.8	14.5					29.8	95.8
55	UNI	0-350	4	4	18.8	41.7	25.8	13.7					29.9	95.8
56	UNI	0-350	5	5	5.6	59.9	11.7	44.1					29.9	95.7
57	UNI	0-560	5	5	7.0	32.4	15.3	25.9					29.7	96.0
58	UNI	0-450	4	4	1.4	48.1	23.3	16.6					29.6	96.3
59	UNI	0-450	5	5	1.3	23.1	25.3	10.1					29.5	96.6
60	UNI				no characteristic direction									
mean				12*	8.8	47.1	10.1	19.6					29.8	95.7

Table 2. Observed and expected directions are presented as the values at the representative locality for each area. The expected direction is computed using the Cretaceous pole (70°N, 208°E, $\alpha_{95} = 5°$) of the China block (Li *et al.* 1988). N is the number of sites, D, I are mean declination and inclination, α_{95} is the confidence circle, k is the precision parameter, λ is the palaeolatitude and VGP is virtual pole position. $R = D$ (observed) $- D$ (expected), $\Delta\lambda = \lambda_{ex} - \lambda_{ob}$. The palaeomagnetic data in Table 2(b) are constructed from our data (*), Achache *et al.* (1984) and Kent *et al.* (1986).

(*a*)

Terrane	Locality			observed					VGP			expected				
Area	Lat. (°)	Long. (°)	N	D(°)	I(°)	α_{95}(°)	K	λ(°)	Lat(°)	Long. (°)	α_{95}(°)	D(°)	I(°)	λ_{ex}(°)	R(°)	$\Delta\lambda$(°)
Sichuan province																
Ya'an	30.1N	103.0E	9	2.1	30.7	11.3	21.7	16.5	76.3N	85.5W	11.1	21.1	40.7	23.2	-19.0	-6.7
Qiangtang Terrane																
Markam	29.7N	98.6E	5	48.2	49.0	8.8	76.9	29.9	48.5N	175.9E	9.5	20.3	38.3	21.5	27.9	8.4
Lhasa Terrane																
Basu	30.1N	96.9E	6	-3.9	23.4	19.2	13.2	12.2	71.8N	70.8W	16.0	20.0	38.1	21.4	-23.9	-9.2

(*b*)

Terrane	(Data source)														
Representative locality				observed (mean)			VGP			expected					
	Lat. (°)	Long. (°)	N	D(°)	I(°)	λ(°)	Lat(°)	Long. (°)	α_{95}(°)	D(°)	I(°)	λ_{ex}(°)	R(°)	$\Delta\lambda$(°)	
Sichuan province	(Ya'an* + Western Sichuan [Kent et al. 1986])														
Ya'an	30.1N	103.0E	10	2.2	32.0	17.4	77.1N	86.3W	9.9	21.1	40.7	23.2	-18.9	-5.8	
Qiangtang Terrane	(Markam*)														
Markam	29.7N	98.6E	5	48.2	49.0	29.9	48.5N	175.9E	9.5	20.3	38.3	21.5	27.9	8.4	
Lhasa Terrane	(Basu* + Lhasa [Achache et al 1984])														
Basu	30.1N	96.9E	20	-8.2	23.1	12.4	58.5N	70.4W	6.0	20.0	38.1	21.4	-28.5	-7.9	

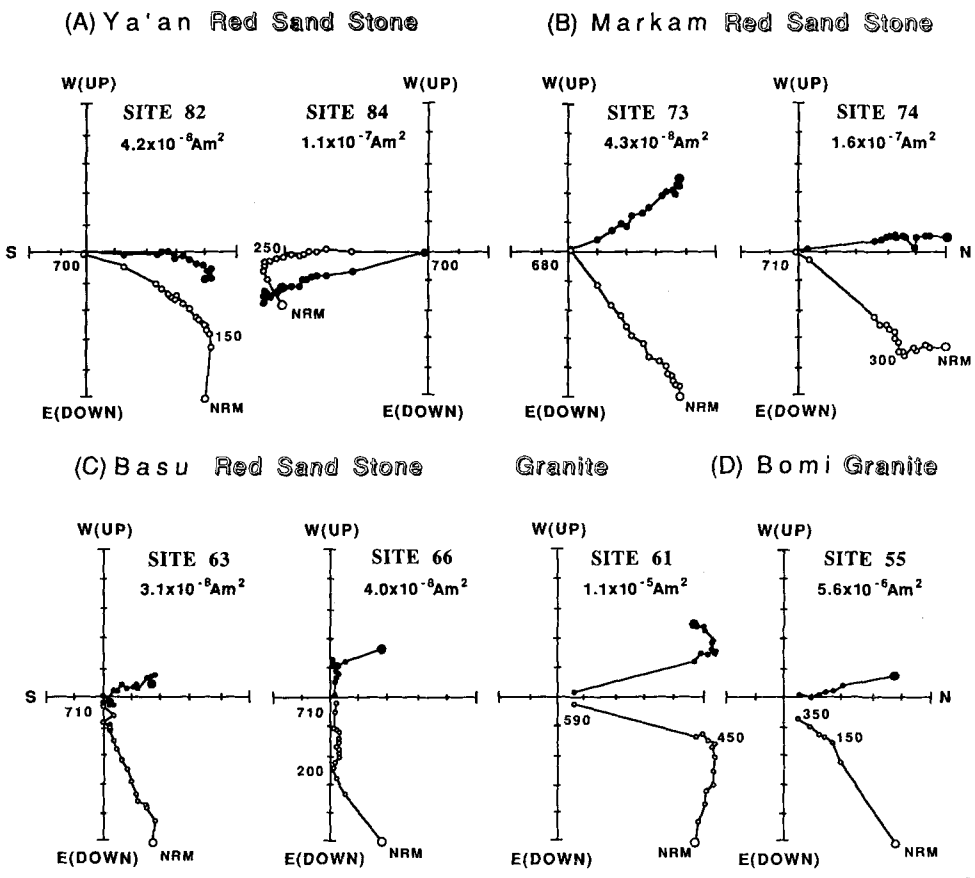


Figure 2. Orthogonal projections of magnetization vector endpoints during thermal demagnetization experiments for specimens. (A) red sandstones from Ya'an, (B) red sandstones from Markam, (C) red sandstones and granitic rocks from Basu and (D) granitic rocks from Bomi. Open (solid) symbols show the magnetic vector projection on to the vertical (horizontal) plane. Numbers by symbols represent temperature (°C). Tectonic corrections have not been applied.

sites after thermal demagnetization above the temperature 530 °C. The HT components probably reside in haematite, because their trajectories converge toward the origin at around 700 °C. The directions of the HT component are composed of both normal and reversed polarities: these directions are antipodal each other. The HT components provide a positive fold test which is significant at the 95 per cent probability level (Fig. 3): the k ratio increased from 9 to 22 for nine sites (McElhinny 1964). The fold test and the presence of three sites with reversed polarity argue for a pre-folding age of magnetization. The mean direction of the HT component directions after tilt correction ($D = 2.1°$, $I = 30.7°$, $\alpha_{95} = 11.3°$) is accepted to be a primary palaeomagnetic direction.

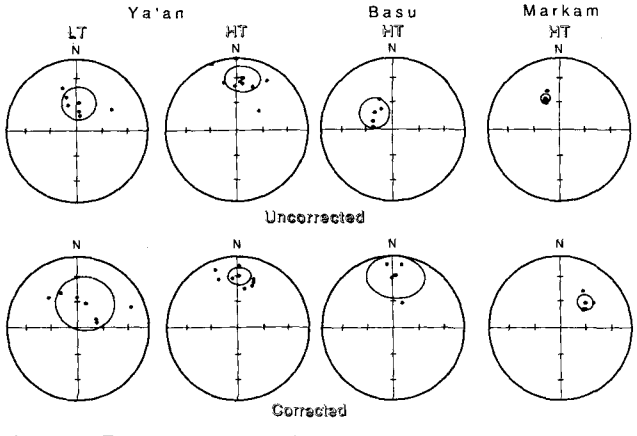


Figure 3. Equal area projection of mean site directions of the high-temperature (HT) components of the red beds at Ya'an, Markam, and Basu areas before and after correcting for bedding (reversed polarity directions are flipped into normal polarity directions). Mean site directions of the low-temperature (LT) components are also drawn for the red beds of the Ya'an area. Circle of 95 per cent confidence about the mean direction is shown. Solid (open) symbols on the lower (upper) hemisphere.

Markam

All five sites of red sandstones have reliable characteristic directions. Two sites are univectorial (Fig. 2). Stable behaviour is also observed during stepwise AF demagnetization, although more than 60 per cent of initial NRM remains up to 95 mT. The other three sites reveal two magnetization components (Fig. 2). The HT component appears above 450 °C. Demagnetization vectors reach the origin at around 700 °C for both the single component and the HT component magnetizations, indicating that their NRMs reside in haematite. The directions of the single component are identical to those of the HT component magnetization. These directions define a characteristic palaeomagnetic direction of Markam.

The Markam area, unfortunately, provides no statistically

decisive fold test, because the Markam sampling sites are located on a homoclinal structure: the bedding variation among sites is less than 13°. However, the characteristic direction is regarded as reliable on the basis of the following facts. The characteristic direction deviates significantly from the present geomagnetic field direction and the present dipole field direction before tilt correction (Fig. 3). As observed in the red sandstones of Ya'an region, the HT components of the Markam region appears after the LT component is eliminated above 450 °C. The characteristic direction after tilt correction ($D = 48.2°$, $I = 49.0°$, $\alpha_{95} = 8.8°$) is used in the final analysis.

Basu

Four out of the five sites of red sandstones have reliable characteristic directions of single component or the HT component magnetizations (Fig. 2). Two sites give a LT component which is eliminated above 350 °C. The LT component falls close to the present geomagnetic directions or present dipole field direction (see Table 1), whereas the characteristic directions of single component or the HT component have a northeast declination with a steep positive inclination and deviate by 30° from the present field (Fig. 3). These facts imply that the LT component is due to post-folding remagnetization and the HT component is possibly a pre-folding one, although the fold test is not applied because we collected red sandstones from a homoclinal structure in Basu. We accept the characteristic directions of the single component and the HT component after tilt correction ($D = 3.3°$, $I = 29.0°$, $\alpha_{95} = 28.0°$) as reliable data for a primary palaeomagnetic direction.

Two sites of granites from the Basu area (site: 61, 62) reveal two components of magnetization (Fig. 2), whereas a demagnetization trajectory of the HT component above 300 °C converges toward the origin. Their NRMs show a restricted blocking temperature range at around 580 °C, indicating that the HT component resides in titanium-poor magnetite. The characteristic directions of the HT component are composed of antipodal normal and reversed polarities (Fig. 4). These directions deviate significantly from the present geomagnetic field direction, and are fairly close to the palaeomagnetic directions of red sandstones after tilt correction. Their characteristic directions ($D = -16.3°$, $I = 12.0°$) are regarded as reliable, although tilt correction is not available.

We use a mean direction ($D = -3.9°$, $I = 23.4°$, $\alpha_{95} = 19.2°$) calculated from both data of red sandstones and granites as a Cretaceous palaeomagnetic direction of Basu in the final analysis.

Bomi

Almost all samples have single component magnetizations. Samples from four sites (site: 47, 50, 55, 56) lose their NRMs, retaining to less than 5 per cent in intensity of the original NRMs, at about 350 °C in thermal demagnetization. Their unblocking temperatures are restricted to narrow range between 300 °C and 350 °C. Low Curie point pyrrhotite could be an important carrier of magnetization of these sites. Samples from the other 10 sites show a broad blocking temperature range between 100 °C and 500 °C (Fig. 2). More than 50 per cent of initial NRM is removed by 250 °C. The low-temperature component is probably due to a thermoviscous remagnetization acquired in the Recent time (Pullaiah *et al.* 1975; Kent 1985).

A total of 12 out of 14 sites of granites give characteristic directions. These directions are of normal polarity and are fairly parallel to the present geomagnetic direction ($D = 8.8°$, $I = 47.1°$, $\alpha_{95} = 10.1°$ versus $D = -1°$, $I = 44°$) (see Fig. 4). These directions are attributed to a secondary origin, probably acquired in the Recent time. It should be noted that the U–Pb method gave ages of 65–95 Myr to granitic rocks of the Bomi area whereas the K–Ar study led to younger ages of 20–38 Myr (So *et al.* 1982). This discrepancy in radiometric ages between the U–Pb and K–Ar methods suggests that the granitic rocks of the Bomi area have experienced argon loss probably due to an increase in temperature after intrusion in the Middle to Late Cretaceous. According to the geological study of Wang & Chu (1988), granitic rocks of the Bomi area have been metamorphosed. These facts imply that the granitic rocks have suffered remagnetization (PTRM or CRM) during some tectonic event accompanied with metamorphism. These directions are, therefore, excluded from further palaeomagnetic investigation.

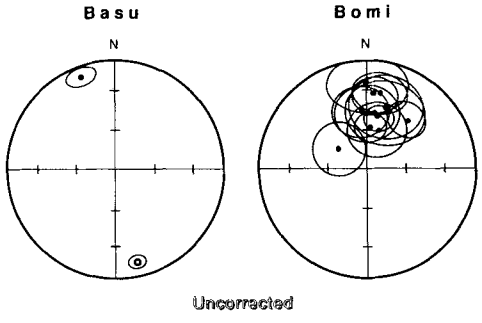


Figure 4. Equal area projection of mean site directions of the high-temperature (HT) components and single components of granitic rocks from Basu and Bomi areas. Tectonic corrections have not been applied. Circle of 95 per cent confidence about each mean site direction is shown. Solid (open) symbols on the lower (upper) hemisphere.

DISCUSSION

Comparison with previous data from Tibet and the Yangtze Terrane

The palaeomagnetic direction ($D = -3.9°$, $I = 23.4°$, $\alpha_{95} = 19.2°$) of the Basu Formation is compared with that of the Albian–Aptian Takena formation exposed in the northern part of the Lhasa Terrane (Achache *et al.* 1984) (see Fig. 5). The expected direction at Basu ($D = -10.4°$ and $I = 21.7°$) is calculated from the data of the Takena formation ($D = -12.3°$, $I = 24.0°$, $\alpha_{95} = 6.6°$) at the northern part of Lhasa. The observed palaeomagnetic directions are compatible with the expected ones at Basu: the difference is $\Delta D = 6.5° \pm 22.2°$, $\Delta I = 1.7° \pm 20.3°$. This demonstrates that the Basu and the Lhasa areas have undergone little relative movement, indicating that the southern part of the Lhasa Terrane including the Lhasa and the Basu areas escapes

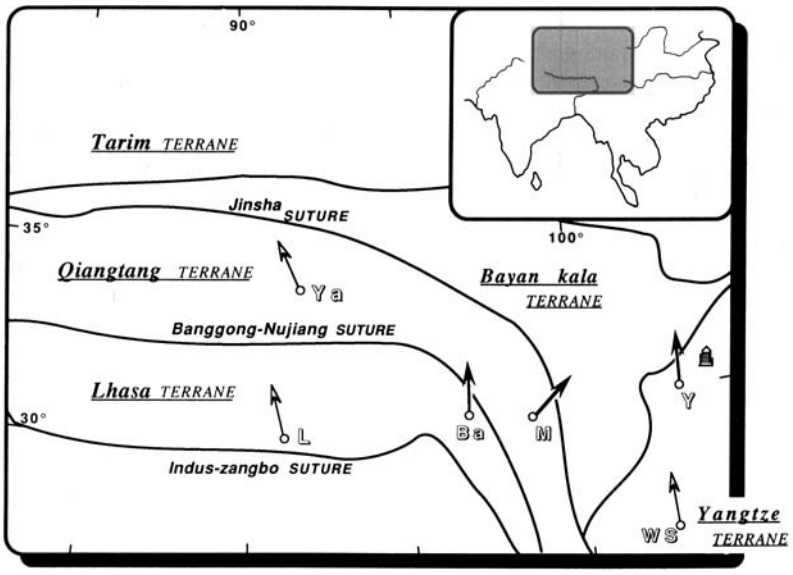


Figure 5. Cretaceous palaeomagnetic directions found in eastern Tibet. The solid arrows are obtained from this study and the open arrows from the previous studies at Yanshiping (Ya) (Lin & Watts 1988), Lhasa (L) (Achache *et al.* 1984) and Western Sichuan (WS) (Kent *et al.* 1986). Ba, Basu; M, Markam; Y, Ya'an. Data at Yanshiping are from Jurassic rocks.

deforming. We, therefore, conclude that the mean of the two data well determine a palaeomagnetic direction for the Lhasa block of the Cretaceous.

Steep inclination values have been reported from Yanshiping of the central part of the Qiangtang Terrane (Lin & Watts 1988) (see Fig. 5), although the palaeomagnetic data were from Jurassic sedimentary rocks. The Yanshiping Group with Bathonian to Kimmeridgian age has $D = 339°$, $I = 58°$ with $\alpha_{95} = 7.2°$. The steep inclination is fairly consistent with that ($I = 49°$) observed in Markam in the eastern part of the Qiangtang Terrane. Palaeomagnetic data of the Qiangtang Terrane are characterized by relatively steep inclination values for the Jurassic and Cretaceous time.

Our data at Ya'an are compared with those of preliminary results obtained by Enkin *et al.* (1988) ($D = 2.1°$, $I = 30.7°$, $\alpha_{95} = 11.3°$ versus $D = 2°$, $I = 35°$, $\alpha_{95} = 9°$). Reliability of palaeomagnetic direction at Ya'an is ascertained through concordance in palaeomagnetic directions of two independent results. Kent *et al.* (1986) reported the Cretaceous palaeomagnetic pole (80.8°N, 296.8°E, $\alpha_{95} = 7.7°$) from Western Sichuan at 26.5°N, 102.3°E (see Fig. 5). The expected direction calculated from this pole is $D = -2.3° \pm 8.3°$, $I = 37.7° \pm 11.0°$ at Ya'an. The difference between observed and expected directions is $\Delta D = 4.4° \pm 15.5°$, $\Delta I = -7.0° \pm 15.8°$. Palaeomagnetic studies thus provide concordant Cretaceous directions within the Sichuan province.

Clockwise rotation of the Qiangtang Terrane in the Three Rivers region

The most striking result from our work is a large clockwise deflection ($D = 48°$) in Cretaceous palaeomagnetic direction in the Qiangtang Terrane (Fig. 5). This clockwise deflection is probably peculiar to Mesozoic palaeomagnetic direction of the Three Rivers region within the Qiangtang Terrane, because central part of the Qiangtang Terrane (the Yanshiping Group) shows fairly northward direction in the Jurassic time (Lin & Watts 1988).

The amount of palaeomagnetic rotation of the Qiangtang Terrane at the Three Rivers region is estimated from comparison with Cretaceous data from the neighbouring blocks. Declination data for the Lhasa Terrane and Sichuan block in our study are $D = -4°$ and $D = 2°$, respectively, whereas the Markam area in the Qiangtang Terrane has $D = 48°$. The amount of rotation is about 50° with respect to two blocks of both sides of the Qiangtang Terrane.

The large clockwise palaeomagnetic rotation of about 50° is consistent with a bending in contour of the suture zones of the terranes in the Three Rivers region (see Figs 1 and 5). The Qiangtang Terrane is bounded by the Bangong–Nujian and Jinsha suture zones. The two suture zones form arcuate shapes which are convex northeastward in eastern Tibet. The strike of the suture zones is nearly N70°W at the longitude of 95°E, whereas the strikes are N30°W for the Bangong–Nujian suture zone and N10°W for the Jinsha suture zones at latitude 30°N. The amount of bending in contour of the suture zones is about 50°, compatible with the palaeomagnetic rotation of the Qiangtang Terrane. Clockwise palaeomagnetic rotation of 50° provides strong geophysical evidence for post-Cretaceous deformation of the terranes in the Three Rivers region.

Our palaeomagnetic data also suggest that curvilineal nature of the large rivers in eastern Tibet is reflection of the tectonic deformation, because the Saluween and Yangtze rivers flow along the suture zones of the Bangong–Nujiang and Jinsha. The large kink shape of the large rivers and the arcuate shape of the orogenic belt in the Three Rivers region was probably formed by the tectonic deformation which took place later than the Cretaceous.

Northward translation of the Sichuan province

The Cretaceous palaeomagnetic pole (77°N, 247°E) constructed from Ya'an and Western Sichuan (Kent *et al.*

1986) is apart from that (76°N, 226°E) of Yangtze block (SCB) (Lin, Fuller & Zhang 1985). This suggests that the Sichuan province behaved as a tectonically different block within the Yangtze Terrane since the Cretaceous.

We compare the difference between two inclination values at Ya'an expected from palaeomagnetic poles of Sichuan province ($I_{Sichuan}$) and the Yangtze Terrane ($I_{Yangtze}$). The difference is $\Delta I = 7°$: $I_{Sichuan} = 32°$ versus $I_{Yangtze} = 39°$. This shows that the Sichuan province had been located at southerly position with respect to the main part of the Yangtze Terrane until the Cretaceous time. The southerly position of the Sichuan block has been inferred from the palaeomagnetic data of the Triassic Telang Formation from the area between Sichuan and Guizhou province (Opdyke *et al.* 1986).

Tectonic evolution of the eastern Tibet

The magnetic field directions from the three blocks (Lhasa, Qiangtang, Sichuan) are compared with the expected directions (Fig. 6 and Table 2b). The Cretaceous palaeomagnetic direction for each block of eastern Tibet has been compiled from our data and previous ones (Achache *et al.* 1984; Kent *et al.* 1986). The expected directions have been computed using the Cretaceous pole (70°N, 208°E. $\alpha_{95} = 5°$) derived from a China block consisting of the Tarim, Yangtze (SCB) and Sino–Korea (NCB) Cratons (Li *et al.* 1988), because these cratons had been united as a single China block until the Cretaceous time (Li *et al.* 1988).

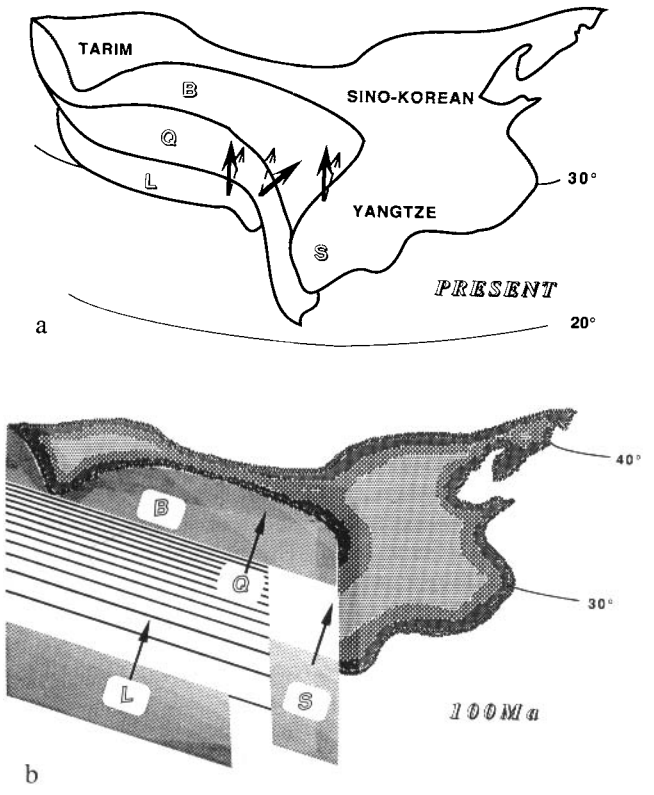


Figure 6. Arrangement in tectonic blocks at the present (A) and the Cretaceous time (B). (A) Present: the solid arrow marks the observed declination and the open arrow with dashed line the Cretaceous expected declination. L, Lhasa; Q, Qiangtang; S, Sichuan; B. Bayan Kala. (B) 100 Ma: three blocks of Lhasa, Qiangtang and Sichuan are superimposed upon the China block with palaeolatitude of the Cretaceous. The palaeolatitude of the China block is calculated from the Cretaceous pole (70°N, 208°E) derived from the stable China block consisting of the Tarim, Yangtze (SCB) and Sino–Korea (NCB) cratons (Li *et al.* 1988). The palaeomagnetic direction lies along the Cretaceous palaeolongitude.

Figure 6(a) shows the relative rotation of the three blocks with respect to the China block since the Cretaceous. While the Qiangtang block has been rotated clockwise through 50° with respect to the Lhasa and the Sichuan blocks in the Three Rivers region, it has experienced clockwise rotation through about 30° with respect to the China block. The Lhasa and Sichuan blocks, on the contrary, experienced the counterclockwise rotation through about 20°. The rotation aspect is summarized as a strong clockwise rotation (more than 30°) of the Three Rivers region near the corner of the Indian continent and a small counterclockwise rotation ($\sim 20°$) of the Lhasa block in front of the collision zone between the India and Eurasia.

Figure 6(b) shows the palaeoposition of the Cretaceous time for the three blocks with respect to the China block. The Qiangtang Terrane has fairly high palaeolatitude (~ 30°N). This terrane has been probably arranged in approximately its present relative position or in a slightly more northerly position with respect to the China block since the Cretaceous time. Our reconstruction places the Lhasa and the Sichuan blocks at southerly latitude: palaeolatitudes of about 12°N and 17°N, respectively. A southerly position of the Lhasa block has been claimed by Westphal *et al.* (1983) and Achache *et al.* (1984) on the basis of the palaeomagnetic data of the Takena Formation.

The deformation process of eastern Tibet is inferred from the arrangement in the tectonic blocks and their relative rotation (Fig. 6). The Lhasa Terrane translated northward by about 1000 km since the Cretaceous, and the Sichuan block has also translated more or less northward (~ 500 km). The clockwise bend in the Qiangtang Terrane is attributed to the post-Cretaceous northward movement of the Lhasa Terrane. The northward motion of the Lhasa Terrane probably began at about 50 Ma when the Indian continent collided with the southern part of Asia (Patriat & Achache 1984; Achache *et al.* 1984; Molnar & Tapponnier 1975). As the indentation of Indian continent progressed into the Asia, continental crust was shortened in the Tibetan region (Achache *et al.* 1984; Besse *et al.* 1984; Chang *et al.* 1986). The Qiangtang Terrane was compressed and shortened in the central part of Tibet, while this Terrane was deformed to form a sharp bend in the eastern part of Tibet where the eastern edge of Indian continent approached.

Deformation due to the collision of India with Asia has been studied by both laboratory and numerical experiments in which a rigid die of India indented a continuum of Asia obeying a power-law rheology (England & McKenzie 1982; Vilotte Daignières & Madariaga 1982; Tapponnier *et al.* 1982; Houseman & England 1986; Cohen & Morgan 1986; Peltzer & Tapponnier 1988). The bending aspect of eastern Tibet has been predicted by both laboratory and numerical experiments. Our observation in rotation mode is only accomplished for the model of England & Houseman (1986) in which an arcuate indenter collides with to the continuum. Our palaeomagnetic results support the idea that the Asian

continent behaved as a continuum of power-law rheology during the collision event and further suggest that the rigid indenter of India had arcuate shape in its northern front.

The motion of the Sichuan block has not been predicted by any models. The Sichuan block behaved as an independent block within the Asian continent. The Sichuan block is possibly an anomalously strong region within a homogeneous rheological material of Asia as the Tarim basin (England & Houseman 1985).

CONCLUSIONS

(1) Palaeomagnetic study provides evidence for tectonic deformation in eastern Tibet.

(2) The Qiangtang Terrane experienced clockwise rotation in the Three Rivers region since the Cretaceous. The amount of the rotation is about 50° with respect to the Lhasa Terrane and Sichuan block.

(3) The large clockwise rotation of the Three Rivers region is attributed to deformation of the eastern part of the Qiangtang Terrane due to a post-Cretaceous northward translation of the Lhasa block associated with the indentation of Indian continent.

(4) The Sichuan province translated northward by about 500 km with respect to the China block since the Cretaceous, accompanied with counterclockwise rotation through about 20°. The Sichuan province was united with the China block later than the Cretaceous.

ACKNOWLEDGMENTS

We are indebted to the members of the Kobe University Scientific and Mountaineering expedition to Tibet 1986 for the courtesy during the field survey. We wish to acknowledge help in the field work from Takashi Okimura. The assistance of Jun Kadoi in the laboratory work is appreciated. We are grateful to Michael W. McElhinny for his review of this paper.

REFERENCES

Achache, J., Courtillot, V. & Xiu, Z. Y., 1984. Paleogeographic and tectonic evolution of southern Tibet since Middle Cretaceous time: New paleomagnetic data and synthesis, *J. geophys. Res.*, **89,** 10 311–10 339.

Allègre, C. J. *et al.*, 1984. Structure and evolution of the Himalaya–Tibet orogenic belt, *Nature,* **307,** 17–22.

Besse, J., Courtillot, V., Pozzi. J. P., Westphal, M. & Zhou, Y. X., 1984. Paleomagnetic estimates of crustal shortening in the Himalaya thrusts and Zangbo suture, *Nature,* **311,** 621–626.

Chang, C. *et al.*, 1986. Preliminary conclusions of the Royal Society and Academia Sinica 1985 geotraverse of Tibet, *Nature,* **323,** 501–507.

Chen, J., Cao, M., Lin, Q. & Liu, X., 1986. Discovery of Jingxing fauna in eastern Tibet, in *Special Issue of Hengduan Mountains scientific expedition (II)—The Comprehensive Scientific Expedition to the Qinghai–Xizang Plateau,* pp. 47–54, ed. The Chinese Academy of Sciences, Beijing Science and Technology Press, Beijing.

Cohen, S. C. & Morgan, R. C., 1986. Intraplate deformation due to continental collisions: a numerical study of deformation in a thin viscous sheet, *Tectonophysics,* **132,** 247–259.

England, P. & McKenzie, D., 1982. A thin viscous sheet model for continental deformation, *Geophys. J. R. astr. Soc.,* **70,** 295–321.

England, P. & Houseman, G., 1985. Role of lithospheric strength heterogeneities in the tectonics of Tibet and neighbouring regions, *Nature,* **315,** 297–301.

England, P. & Houseman, G., 1986. Finite strain calculations of continental deformation: 2. comparison with the India–Asia collision zone, *J. geophys. Res.,* **91,** 3664–3676.

Enkin, R. J., Chen, Y., Courtillot, V., Besse, J., Xing, L., Zhuang, Z., Zhang, J., Zhang, Z. & Xu, X., 1988. Paleomagnetic results from a Mesozoic red bed sequence in the Sichuan Basin, China, *EOS, Trans. Am. geophys, Un.,* **69,** 338.

Geological Map of the Tibetan Plateau, 1981. 1/1 500 000, Chendu, China.

Hao, Y. *et al.*, 1986. *Stratigraphy of China No. 12—The Cretaceous System of China,* Geological Publishing House, Beijing.

Houseman, G. & England, P., 1986. Finite strain calculations of continental deformation: 1. Method and general results for convergent zones, *J. geophys. Res.,* **91,** 3651–3663.

IAGA Division I Working Group 1, 1987. International geomagnetic reference field revision 1987, *J. Geomag. Geoelectr.,* **39,** 773–779.

Kent, D. V., 1985. Thermoviscous remagnetization in some Appalachian limestones, *Geophys. Res. Lett.,* **12,** 805–808.

Kent, D. V., Xu, G., Huang, K., Wen-You, Z. & Opdyke, N. D., 1986. Paleomagnetism of upper Cretaceous rocks from South China, *Earth planet. Sci. Lett.,* **79,** 179–184.

Kirschvink, J. L., 1980, The least-square line and plane and the analysis of palaeomagnetic data, *Geophys, J. R. astr. Soc.,* **62,** 699–718.

Li, Y., Zhang, Z., McWilliams, M., Sharps, R., Zhai, Y., Li, Y. G., Li, Q. & Cox, A., 1988. Mesozoic paleomagnetic results of the Tarim Craton: Tertiary relative motion between China and Siberia, *Geophys. Res. Lett.,* **15,** 217–220.

Lin, J. L. & Watts, D. R., 1988. Paleomagnetic results from the Tibetan Plateau, *Phil. Trans. R. Soc. Lond.,* A, **327,** 239–262.

Lin, J. L., Fuller, M. & Zhang, W. Y., 1985. Preliminary Phanerozoic polar wander paths for the North and South China blocks, *Nature,* **313,** 444–449.

McElhinny, M. W., 1964. Statistical significance of the fold test in palaeomagnetism, *Geophys. J. R. astr. Soc.,* **8,** 338–340.

Molnar, P. & Tapponnier, P., 1975. Cenozoic tectonics of Asia: Effects of a continental collision, *Science,* **189,** 419–426.

Molnar, P. & Tapponnier, P., 1978. Active tectonics of Tibet, *J. geophys. Res.,* **83,** 5361–5375.

Opdyke, N. D., Huang, K., Xu, G., Zhang, W. Y. & Kent, D. V., 1986. Paleomagnetic results from the Triassic of the Yangtze Platform, *J. geophys. Res.,* **91,** 9553–9568.

Patriat, P. & Achache, J., 1984. India–Eurasia collision chronology has implications for crustal shortening and driving mechanism of plates, *Nature,* **311,** 615–621.

Peltzer, G. & Tapponnier, P., 1988. Formation and evolution of strike–slip faults, rifts and basins during the India–Asia collision: An experimental approach, *J. geophys. Res.,* **93,** 15 085–15 117.

Pullaiah, G., Irving, E., Buchan, K. L. & Dunlop, D. J., 1975. Magnetization changes caused by burial and uplift, *Earth planet. Sci. Lett.,* **28,** 133–143.

Sichuan Institute of Geology and Mineral Resources, 1982. *Continental Mesozoic Stratigraphy and Paleontology in Sichuan Basin of China,* People's Publishing House of Sichuan, Chendu.

So, K. *et al.*, 1982. Geochemistry of granitoid rocks in southern Xizang, *The Series of the Scientific Expedition to Qinghai–Xizang Plateau,* Scientific Publishing House, Beijing.

Tapponnier, P., Peltzer, G., Le Dain, A. Y., Armijo, R. & Cobbold, P., 1982. Propagating extrusion tectonics in Asia: New insights from simple experiments with plasticine, *Geology,* **10,** 611–616.

Vilotte, J. P., Daignières, M. & Madariaga, R., 1982. Numerical

modeling of intraplate deformation: Simple mechanical models of continental collision, *J. geophys. Res.*, **87,** 10 709–10 728.

Wang, E. & Chu, J. J., 1988. Collision tectonics in the Cenozoic orogenic zone bordering China, India and Burma, *Tectonophysics,* **147,** 71–84.

Wei, Z. & Tan, Y., 1979. An outline of the stratigraphy in Xizang (Tibet), in *Contribution to the Geology of Qinghai–Xizang (Tibet) Plateau,* pp. 1–38, ed. CGQXP Editorial Committee, Ministry of Geology and Mineral Resources, Geological Publishing House, Beijing.

Westphal, M., Pozzi, J. P., Zhou, Y. X., Xing, L. S. & Chen, X. Y., 1983. Paleomagnetic data about southern Tibet (Xizang)—I. The Cretaceous formations of the Lhasa block, *Geophys. J. R. astr. Soc.,* **73,** 507–521.

Zhang, Z. H., Liou, J. G. & Coleman, R. G., 1984. An outline of the plate tectonics of China, *Geol. Soc. Am. Bull.,* **95,** 295–312.

Zhao, X. X. & Coe, R. S., 1987. Palaeomagnetic constraints on the collision and rotation of North and South China, *Nature,* **327,** 141–144.

Zijderveld, J. D. A., 1967. A.C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism,* pp. 254–286, eds Collinson, D. W., Creer, K. M. & Runcorn, S. K., Elsevier, Amsterdam.